

exmp:

$$1. \forall x (P(x) \Rightarrow Q(x))$$

Hyp.

$$2. \exists x (R(x) \wedge \forall y (Q(y) \Rightarrow \neg S(x, y)))$$

Hyp.

$$3. R(k) \wedge \forall y (Q(y) \Rightarrow \neg S(k, y))$$

EI(2) $x \rightarrow k$: constant

$$4. \forall y (Q(y) \Rightarrow \neg S(k, y))$$

Simp(3)

$$5. Q(y) \Rightarrow \neg S(k, y)$$

UI(4) $y \rightarrow y$: variable

$$6. P(y) \Rightarrow Q(y)$$

UI(1) $x \rightarrow y$: variable
reuse y

$$7. P(y) \Rightarrow \neg S(k, y)$$

H.S. (5) + (6)

$$8. \forall y (P(y) \Rightarrow \neg S(k, y))$$

UG(7)

$$9. R(k)$$

Simp(3)

$$10. R(k) \wedge \forall y (P(y) \Rightarrow \neg S(k, y)) \quad \text{Conj (8) + (9)}$$

$$11. \exists x (R(x) \wedge \forall y (P(y) \Rightarrow \neg S(x, y))) \quad \text{EG(10)}$$

$$\exists x (R(x) \wedge \forall y (P(y) \Rightarrow \neg S(x, y))) \quad \text{Conclusion}$$

H.W.
H1 - Every student passes at least one of courses they take.

H2 - Some student dislikes all courses they take

C - Some student passes some courses they don't like.

$U = \{ \text{all students and all courses} \}$

$S(x)$: x is student

$C(y)$: y is course

$T(x,y)$: student x takes course y

$P(x,y)$: student x passes course y

$L(x,y)$: student x likes course y

1. $\forall x (S(x) \Rightarrow \exists y (C(y) \wedge T(x,y) \wedge P(x,y)))$ H1

2. $\exists x (S(x) \wedge \forall y (C(y) \wedge T(x,y) \Rightarrow \sim L(x,y)))$ H2

⋮

C. $\exists x (S(x) \wedge \exists y (C(y) \wedge T(x,y) \wedge P(x,y) \wedge \sim L(x,y)))$

exmp:

1. $\forall x (P(x) \Rightarrow Q(x))$
2. $\exists x (R(x) \wedge \sim Q(x))$
3. $\forall z (R(z) \Rightarrow P(z) \vee S(z))$
4. $R(t) \wedge \sim Q(t)$
5. $R(t)$
6. $\sim Q(t)$
7. $P(t) \Rightarrow Q(t)$
8. $\sim P(t)$
9. $R(t) \Rightarrow P(t) \vee S(t)$
10. $P(t) \vee S(t)$
11. $\sim \sim P(t) \vee S(t)$
12. $\sim P(t) \Rightarrow S(t)$
13. $S(t)$
14. $R(t) \wedge S(t)$
15. $\exists x (R(x) \wedge S(x))$

Conclusion:

$$\exists x (R(x) \wedge S(x))$$

EI ② t : constant

Simp ④

Simp ④

UI ① t : reuse

M.T. ⑥ + ⑦

UI ③ t : reuse

M.P. ⑤ + ⑨

double neg. ⑩

Imp ⑪

M.P. ⑧ + ⑫

Conj ⑬ + ⑬

EG ⑭

$$A \Rightarrow B \equiv \sim A \vee B$$

Set: collection of items

$$A = \{1, 3, 1, 2, 3, \text{dog}\} = \{2, 1, 3, \text{dog}\}$$

$$|A| = 4 \quad 3 \in A \quad 7 \notin A$$

$$\mathbb{Z}: \{\dots, -2, -1, 0, 1, 2, \dots\}$$

$$\mathbb{N}: \{1, 2, 3, \dots\} = \mathbb{Z}^+$$

$$\mathbb{Z}^-: \{\dots, -3, -2, -1\}$$

$$\mathbb{Q}: \left\{ \frac{p}{q} : p \in \mathbb{Z} \wedge q \in \mathbb{Z} \wedge q \neq 0 \right\} \text{ rational}$$

$$2.5 = \frac{5}{2} \quad 0.25 = \frac{1}{4}$$

$$\sqrt{2} \notin \mathbb{Q} \quad \pi \notin \mathbb{Q}$$

$$\mathbb{R}: \text{real numbers} \quad \sqrt{2} \in \mathbb{R}, \pi \in \mathbb{R}$$

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$$

Power set of A , $P(A)$, 2^A : the set of all subsets of A

$$A = \{1, 2, 3\} \text{ what is } P(A) = ?$$

$$P(A) = \{ \{ \}, \{1\}, \{2\}, \{3\}, \{2, 3\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\} \}$$

$$|A| = 3 \rightarrow |P(A)| = 8 = 2^3$$

$$\text{if } |A| = n \text{ then } |P(A)| = 2^n$$

Cartesian product of A, B

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

$(a, b) \neq (b, a)$
↓
ordered pair

$$A = \{1, 2, 3\}, B = \{p, q\}$$

$$A \times B = \{(1, p), (1, q), (2, p), (2, q), (3, p), (3, q)\} \rightarrow (p, 2) \notin A \times B$$

$(2, p) \in A \times B$

$$|A \times B| = |A| \times |B| = 3 \times 2 = 6$$

$$B \times A = \{(p, 1), (p, 2), (p, 3), (q, 1), (q, 2), (q, 3)\} \rightarrow (p, 2) \in B \times A$$

$$A \times B \neq B \times A$$

$$|A \times B| = |B \times A|$$

A binary relation between A and B is a subset of $A \times B$

$$R_1 = \{(1, p), (1, q), (2, p)\}$$

↑
relation

$$R_2 = \{(2, q)\}$$

→ $(1, q) \in R_1$

1 $R_1 q$

2 $R_1 p$

3 $R_1 q$

2 $R_2 q$

1 $R_2 q$